Probabilistic Prediction of Outcome and Herding Time in Single-Pursuer Multi-Evader Herding

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***Abstract*—A single pursuer can herd many non-cooperative evaders by circling a goal with a radius that follows one tracked evader. Existing theory for this law is local: it guarantees a stable herd near the equilibrium but says nothing about which starting configurations end in success, in a stalled herd, or in a lost evader, or how long herding takes. We learn this map from 60,000 validated simulations with classical, probabilistic machine learning. A Gaussian-process (GP) classifier on permutation-invariant features predicts the three-way outcome from the start alone with 93.7 % balanced accuracy and the best calibrated probabilities of all models, and reveals that evaders are lost even where the stability condition of the theory holds; a set kernel keeps 93.8 % on flock sizes never seen in training. A kernel mixture-cure survival model separates slow from stuck herds beyond the simulation horizon (AUC 0.98 against ten-times-longer re-simulation). A risk-aware active-learning rule that picks the next simulation by the Bayes risk of the decision reaches the decision cost of random sampling with 2.3× fewer simulations, whereas entropy sampling worsens it. Used to choose the pursuer settings, the risk-aware model matches the theory-guided rule's zero losses while herding 2.3× faster in simulation.**

I. Introduction

Herding many evaders with one pursuer is a hard nonlinear, switched-like control problem. Singh and Chakraborty proposed a circular pursuer trajectory [1] and a feedback variant whose radius follows the evader that was farthest at the start [2]. Their theorems give existence, uniqueness and local asymptotic stability of a common equilibrium ring under a parameter condition, and an inner (Lyapunov-ellipse) estimate of the region of attraction for the tracked evader. For the other evaders, the region is only shown by brute-force simulation [2].

A practitioner needs more: *given this flock and these settings, will herding succeed, stall, or lose an evader, how sure are we, and how long will it take?* We treat this as a pattern-recognition problem. Contributions:

1) A validated dataset and calibrated probabilistic predictors (GP classification and regression, Bayes-risk decisions) of the outcome and herding time for any flock size, with an analysis of where the local theory fails.
2) A kernel mixture-cure survival model for herding time with right-censoring, which separates slow herds from herds that never settle.
3) Risk-aware active learning: choosing the next simulation by the Bayes risk of the downstream decision, and the finding that the usual uncertainty-based choice optimises the wrong target.
4) A set kernel (kernel mean embedding) over the evaders that generalises to unseen flock sizes, and a risk-aware choice of pursuer settings validated by simulation against a theory-guided rule.

GP classification with active learning has estimated regions of attraction of single systems [3] and missile capture regions [4]; see also safe ROA learning [5]. We add three asymmetric-cost outcome classes, many agents with permutation-invariant inputs, censored times, and a decision-driven acquisition rule.

II. Related Work

Herding. The circular law [1] yields a stable and an unstable limit cycle and an inner Lyapunov estimate of the region of attraction (ROA); the feedback law [2] adds a unique, locally stable common equilibrium for all evaders when $2k_1^2R^2 > 1$ and $\kappa < \ln(2k_1^2R^2)/(2k_1)$. Global behaviour, losses and herding time are not characterised. *Learning ROAs.* GP classification estimates ROAs with confidence and active learning [3], and missile capture regions with uncertainty sampling [4]; GP-based safe exploration learns ROAs of uncertain systems [5]. Classical Lyapunov estimates [6], [7] are sound but can be conservative; we quantify how much. *Survival models.* Tobit regression [8] handles censoring; mixture-cure models [9] add a sub-population that never experiences the event. We use kernel versions via random features [10]. *Active learning* [11] usually targets uncertainty; we target the Bayes risk of the decision.

III. System and Data

A. Model

Evader i at $x_i \in \mathbb{R}^2$ moves away from the pursuer p with speed k/d_i^2, $d_i = \|x_i - p\|$; evaders do not interact [1]. The pursuer moves at angular speed ω with radius $R_p(t) = R\,e^{k_1(r_0(t)-\kappa)}$, where r_0 is the distance of the tracked evader (farthest at $t = 0$) and $\kappa = r_0(0)$ [2]; $k_1 = 0$ gives the fixed circle of [1] (our baseline law). In the frame rotating with the pursuer the dynamics are autonomous. Scaling lengths by $1/k_1$ and time by $1/\omega$ leaves two parameters, $q = k_1R$ and $\beta = kk_1^3/\omega$, plus the scaled start; for the circle law only $\gamma = k/(\omega R^3)$ remains. The theory's condition is $a = 2q^2e^{-2\kappa} > 1$ (in scaled units).

B. Labels

A run is *loss* if an evader leaves the pursuer circle at $T_{\max} = 300$ laps or goes beyond an escape radius; *success* if, from some lap t_s on, every evader stays within 1 % (radius) and 0.05 rad of the common equilibrium; otherwise *deadlock*. The herding time is $T_h = t_s$.

C. Simulation and data

We wrote our own simulator (RK4 with a step that shrinks during close passes). It reproduces the published ring radii [1], the equilibrium of the running example of [2], an independent loss count for the tracked-index law, and agrees with an adaptive DOP853 solver: 700 of 700 re-simulated runs get the same label and herding time. We generated 60,000 arXiv-law runs ($N = 1..6$, three strata of a: $a > 1$, $6/e^2 < a \le 1$, $a \le 6/e^2$; Latin-hypercube sampling [12]; plus boundary sets) and 10,000 circle-law runs: 26,454 success, 16,013 deadlock, 17,533 loss. A fixed stratified 20 % test set is used throughout.

D. Features

Only the start is used. Lengths are divided by κ. Global: N, $\log q$, $\log \beta$, κ, $\log a$, $\log(q/\kappa)$, $\log(r^*/\kappa)$. Flock: means, spreads and minima of radii, circular variance, largest angular gap, pairwise distances, hull area, angles relative to the pursuer and to the tracked evader, distances to the pursuer and to the equilibrium point. All are permutation-invariant and defined for every N (24 features).

IV. Methods

A. Classifiers and calibration

Baselines: majority class, the theory rule (success iff $a > 1$), logistic regression, and an RBF-SVM (with Platt scaling [13] for probabilities). The GP classifier uses one binary GP per class (Laplace approximation, one-vs-rest [14]) with an ARD kernel $\sigma_f^2 \exp(-\frac{1}{2}\sum_j (x_j - x'_j)^2/\ell_j^2)$. Exact GP cost is $O(n^3)$, so the kernel is tuned by marginal likelihood on 1,000 runs and the GP is fitted on 4,000. We cap $\sigma_f^2 \le 100$: without the cap, three of six random training sets produced an over-confident GP (balanced accuracy down to 0.885); with it, seven sets give 0.936–0.938.

B. Bayes-risk decisions

With cost matrix L (calling a true loss "success" costs 10; loss→deadlock and deadlock→success cost 3; other errors 1), the decision is $\hat{a}(x) = \arg\min_j \sum_i P(i \mid x)\, L_{ij}$.

C. Herding time with censoring

Let $y = \log(1 + T_h)$ and $c = \log(1 + T_{\max})$. With random Fourier features $z(x)$ [10] of the ARD kernel, $f(x) = b + w^\top z(x)$ approximates a GP. The Tobit model [8] uses $y \sim \mathcal{N}(f, \sigma^2)$ for settled runs and $P(y > c) = \Phi((f - c)/\sigma)$ for deadlocks. Our *kernel mixture-cure* model [9] adds $\pi(x) = \mathrm{sigmoid}(g(x))$, the probability that a run ever settles:

$$\mathcal{L} = \prod_{\text{settled}} \pi\, \mathcal{N}(y; f, \sigma^2) \prod_{\text{deadlock}} \left[(1 - \pi) + \pi\, \Phi(\tfrac{f - c}{\sigma})\right], \quad (1)$$

with Gaussian priors on the feature weights (MAP by L-BFGS with analytic gradients).

D. Risk-aware active learning

Starting from 200 random runs, the GP repeatedly adds batches from a pool. *Entropy* sampling picks the largest predictive entropy [11]. *Risk-aware* sampling picks the largest Bayes risk of the best decision, $r(x) = \min_j \sum_i P(i \mid x) L_{ij}$, i.e. where acting is most expensive.

TABLE I
Test-set results (12,000 runs). Bal. acc. = mean per-class recall.

model (training runs)	bal. acc.	log-loss	Brier	ECE
theory rule ($a > 1$)	0.457	–	–	–
logistic regression (48k)	0.825	0.482	0.258	0.039
SVM + Platt (4k)	0.918	0.254	0.130	0.034
SVM + Platt (20k)	0.938	0.200	0.099	0.021
GP classifier (4k)	0.937	**0.162**	**0.092**	0.028

E. Evaluation protocol

All models share one fixed, stratified (label $\times N$) 20 % test set (12,000 runs) that is never used for tuning; settings are chosen by cross-validation on training data only. We report balanced accuracy (mean per-class recall), log-loss, Brier score and expected calibration error (ECE, 15 bins); for times, median error and interval coverage; for censored times, the time-dependent concordance C_{td} [15]. Paired comparisons on the same flocks use exact McNemar and Wilcoxon signed-rank tests. Every learned conclusion that matters is checked by new simulations.

V. Results

A. Outcome prediction

Table I: the GP matches the SVM trained on five times more data in accuracy and gives the best probabilities. Trained on $N \le 4$ and tested on $N = 5, 6$, the SVM keeps 92.8 %. The theory rule reaches only 45.7 %: 52 % of successful runs violate $a > 1$ and 11 % of lost runs satisfy it. Uncertainty is informative: answering only the 80 % most confident runs raises GP accuracy to 98.6 %. The Bayes rule cuts "loss called success" from 169 to 9 test runs. Fig. 1 compares the learned map with 1,600 fresh simulations (97.3 % agreement): losses occur right of $a = 1$, where the theorems hold, because those theorems are local.

B. Herding time

On settled test runs a GP regression predicts T_h with median error 0.7 laps and 89 % coverage of its 90 % bands, but it ignores the 27 % of runs that stop unsettled at $T_{\max}$. Table II compares three censoring-aware variants sharing the same RFF kernel (1,000 features). The mixture-cure model is best on every test measure. The decisive check goes beyond the training horizon: 300 test deadlock runs re-simulated $10\times$ longer (3,000 laps) split into 158 that settle, 137 that never do and 5 losses. From the start alone the cure model predicts which ones settle with AUC 0.984 and Brier 0.049 (Tobit 0.216, successes-only 0.345); its mean predicted probability, 0.535, matches the observed 0.536, and it predicts their herding times within a median factor of 1.09 (Fig. 2). "Deadlock" thus hides two regimes, slow and stuck, which the model separates.

C. Unseen flock sizes: a set kernel

Hand-made summaries lose information about individual evaders. We also compare runs directly

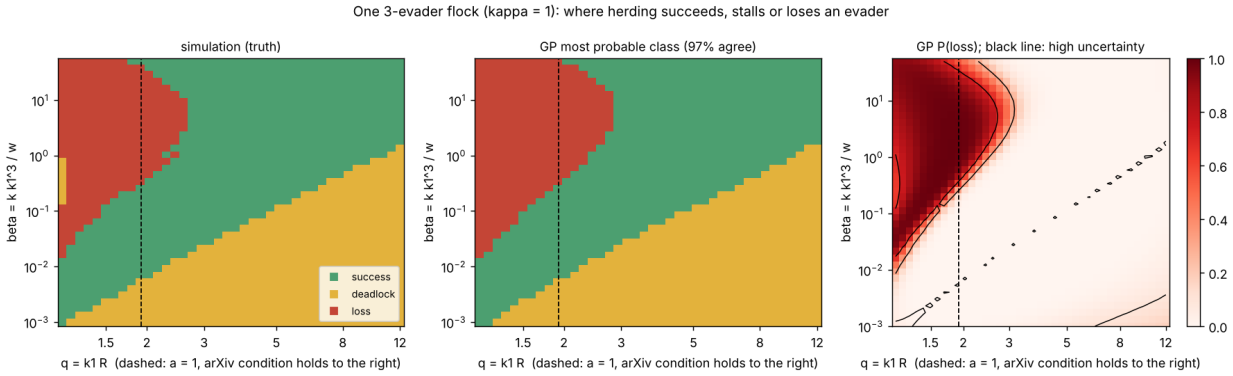


Fig. 1. One 3-evader flock: simulated outcome (left), GP prediction (middle) and GP $P(\text{loss})$ with its high-uncertainty contour (right) over q and β; dashed: $a = 1$.

TABLE II
CENSORED HERDING TIME ON THE TEST SET (SETTLED AND DEADLOCK RUNS).

	successes only	Tobit	cure
log-likelihood / run	-24.05	-0.27	$\mathbf{+0.37}$
time concordance C_{td} [15]	0.943	0.979	**0.992**
settled: median error (laps)	0.72	1.95	**0.69**
deadlock: mean $P(T_h > T_{\max})$	0.83	0.95	**0.98**

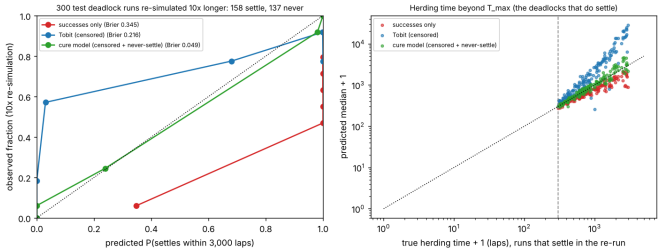


Fig. 2. Deadlock runs re-simulated $10\times$ longer. Left: reliability of $P(\text{settles within 3,000 laps})$. Right: predicted vs true herding time of those that settle.

as sets with $K(A,B) = k_g(g_A, g_B)\,[k_e(e_{0A}, e_{0B}) + \frac{1}{|S_A||S_B|}\sum_{u\in S_A, v\in S_B} k_e(u,v)]$, where g are six global parameters, e_0 the tracked evader and S the set of other evaders (positions/κ); the bracket is an inner product of kernel mean embeddings [16], valid for any set sizes. With an SVM and 8,000 training runs: random split 93.6 % (hand features 93.0 %); trained on $N \le 4$, tested on $N = 5, 6$: **93.8 %** (hand features 90.0 %). The set kernel does not degrade on larger flocks.

D. Unsupervised regimes

PCA and a GMM (BIC) on start features do not recover the classes (ARI 0.08), but on trajectory shapes the clusters are 95 % pure and the first two components form one arm per class (Fig. 3): the labels are natural regimes, and predicting them from the start requires supervision.

TABLE III
SIMULATED OUTCOME OF THE SETTINGS CHOSEN BY EACH STRATEGY (300 FLOCKS).

strategy	success	deadlock	loss	median T_h
random (dataset)	38.0	31.0	31.0	18.5
fixed default	61.7	9.0	29.3	37
theory ($a = 4$)	83.3	16.7	**0.0**	15
GP, max $P(\text{success})$	**86.7**	10.3	3.0	6
GP-safe, $P(\text{loss}) \le 1\%$	81.7	18.3	**0.0**	**6**

E. Risk-aware active learning

Fig. 4: risk-aware sampling reaches the decision cost of random sampling at 4,000 runs with 1,750 ($2.3\times$ fewer) and ends 22 % lower (0.099 vs 0.127). Entropy sampling reaches random's accuracy with 1,500 runs and ends best in accuracy (0.946), yet its decision cost is *worse* than random (0.137): what to simulate should follow the decision the model is for.

F. Using the model to choose the pursuer settings

For 300 held-out multi-evader flocks we let each strategy choose (q, β) on a 15×15 grid and simulate the choice (Table III). A theory-guided rule (choose q so that $a = 4$, $\beta = 1$) never loses an evader. Maximising the GP's $P(\text{success})$ gives more successes (exact McNemar $p = 0.002$) but loses evaders in 3 % of flocks ($p = 0.004$). The risk-aware choice—maximise $P(\text{success})$ subject to $P(\text{loss}) \le 1\,\%$, then the shortest predicted time from the cure model—matches the theory's zero losses (success 81.7 vs 83.3 %, $p = 0.06$) and herds faster: on the 245 flocks where both succeed, median 6 vs 14 laps, faster in 88 % (Wilcoxon $p = 9 \times 10^{-24}$).

G. Theory comparison

For the circle law we re-solve the paper's Lyapunov-ellipse problem [1], [7] (start from the Lyapunov-equation solution, optimise the shape, check $\dot{V} < 0$ on dense grids) and compare it with the true region from a 90×90 grid of simulated starts. The certificate is always correct (no false positives) but covers only 0.5 % ($\gamma = 0.01$) to 3.7 % ($\gamma = 0.3$) of the truly converging starts (Fig. 6). The reason is a nearly undamped

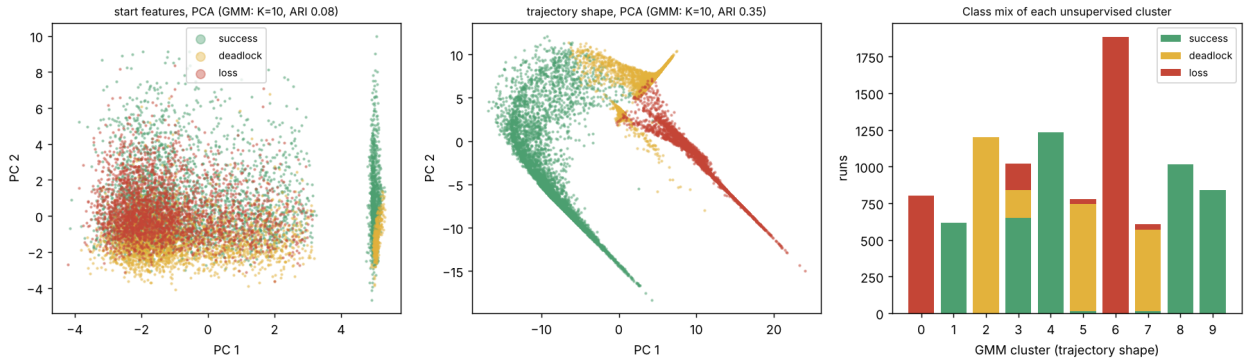


Fig. 3. PCA of start features (left) and of trajectory shapes (middle), coloured by outcome; class mix of the GMM clusters of trajectory shapes (right).

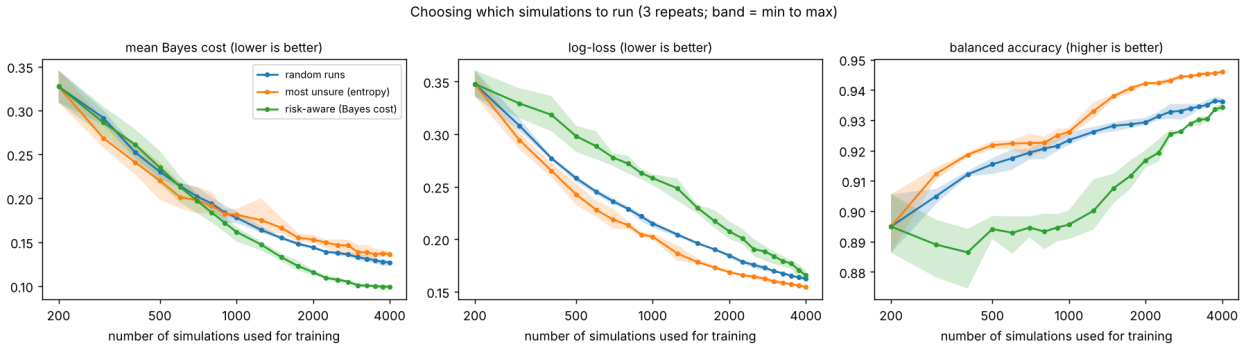


Fig. 4. Quality vs number of simulations for three acquisition rules (3 repeats; bands = min to max).

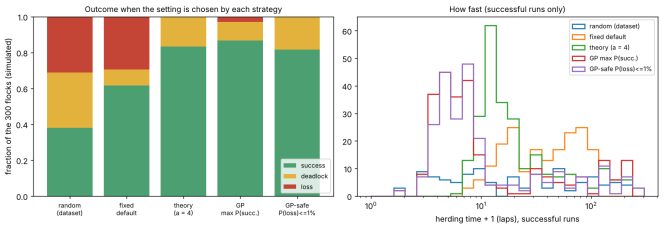


Fig. 5. Outcome (left) and herding time (right) of the settings chosen by each strategy, every choice simulated.

rotation at the ring (Jacobian eigenvalues $\approx -0.005 \pm 1i$ at $\gamma = 0.01$): a quadratic Lyapunov function stops decreasing very close to it. Our certified ellipse is smaller than the one drawn in [1]; along its limiting direction $\dot{V}$ indeed turns positive at $0.08R$. For the arXiv law, Fig. 1 and the theory-rule baseline show the analogous gap.

VI. Conclusion

The outcome and duration of single-pursuer herding are predictable from the start with calibrated uncertainty, far beyond what the local theory certifies: the proven region covers a few percent of the true one, and evaders are lost where the stability condition holds. Two lessons carry beyond herding. First, censored outcomes hide structure: a cure model turns "not finished" into "slow" versus "stuck" and predicts it beyond the horizon. Second, both data collection and decisions should be driven by the cost of the worst mistake: risk-aware active learning needs $2.3\times$ fewer simulations for the same decision quality, and risk-aware setting selection keeps zero losses while herding faster. Limits: all results are in simulation; labels depend on $T_{\max}$ and tolerances; exact GPs were limited to 4,000 training runs; the cost matrix is a design choice. Future work: switching pursuer laws and noisy sensing.

Acknowledgment and AI use

An AI assistant (Claude Code) was used, as allowed in the course, for code, experiments and drafting under the author's direction; all numbers come from the scripts in the project repository and every choice was checked by the author.

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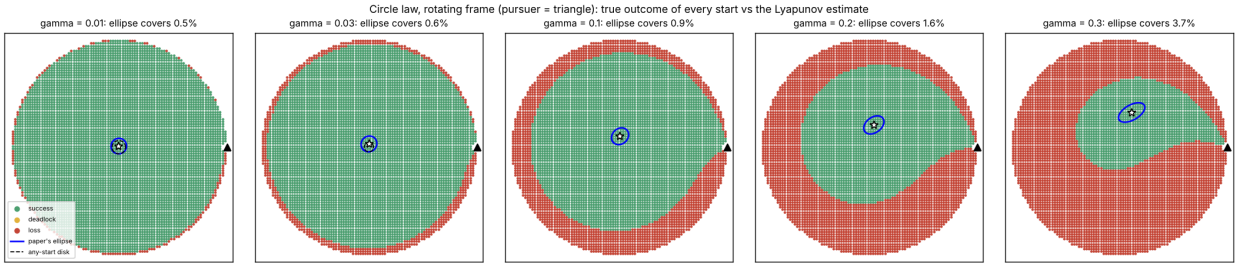


Fig. 6. Circle law, rotating frame: simulated outcome of every start (green: converges, red: lost) and the Lyapunov-ellipse certificate (blue).

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